

# Strain engineering in GaN layers grown on silicon by molecular beam epitaxy: The critical role of growth temperature

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## ABSTRACT

In this work we study both the structural and electrical qualities of AlGaIn/GaN high electron mobility transistor heterostructures grown on silicon(111) by molecular beam epitaxy. Correlations are established between the quality of the structures and the relaxation rate of the mismatch stress in layers grown using ammonia as a nitrogen source. Comparison with layers grown using a nitrogen plasma source confirms the primordial role of the growth temperature for stress relaxation and dislocation filtering.

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## 1. Introduction

The growth of Ga(Al)N films on silicon substrates is difficult due to the difference in thermal expansion coefficient (TEC), which induces a tensile stress in the films at room temperature that can generate cracks. Furthermore, the stressed epitaxial layers and the substrate suffer a bending depending on sign and intensity of the stress. Excessive bending and the presence of cracks hamper the development of devices especially for large substrates for which a significant bowing can appear. The growth of GaN on silicon is particularly challenging because of the large lattice mismatch (17%) but also the difference in TEC between GaN and Si that is estimated to be  $\alpha_{\text{GaN}} - \alpha_{\text{Si}(111)} = 2.9 \times 10^{-6} \text{ K}^{-1}$ . The thermal mismatch is responsible for a tensile strain of 0.23% at room temperature for a totally relaxed GaN film grown at 800 °C. As a consequence, without stress engineering schemes, no thick, high-quality GaN layer can be obtained without cracks on Si substrates. The use of AlN interlayers or AlGaIn stress mitigating layers enables the growth of thick high-quality GaN films on silicon [1–4]. Indeed, the 2.5% lattice mismatch between GaN and AlN allows GaN to be grown compressively strained; even if a major part of this strain is relaxed, a residual stress remains capable of compensating the thermal mismatch stress upon cooling down. Nevertheless, the efficiency of such a scheme

depends much on growth conditions and preceding layer quality. As shown in a previous work [5], *in situ* monitoring of stress is a very useful tool for optimizing such layers. In particular, measurement of the curvature of the silicon wafer during the molecular beam epitaxy (MBE) growth of AlN and GaN layers provides very useful information on strain relaxation for the design of structures based on thick GaN buffer layers such as high electron mobility transistors (HEMTs). In the present work, we extend the investigations and establish correlations between wafer curvature and both structural and electrical qualities of the layers. Furthermore, we show that the growth temperature is a critical parameter that strongly influences the inclination of dislocations and their annihilation in the GaN buffer layer.

## 2. Experimental procedure

In the present work, AlGaIn/GaN heterostructures have been grown in a molecular beam epitaxy system (Riber Compact 21) equipped with both ammonia [6] and nitrogen plasma sources. Liquid nitrogen cryo-panels as well as turbo-molecular pumping (2800 l/s) are used to produce the high vacuum conditions necessary for the growth of high-purity materials. An ammonia flow rate of typically 100–200 sccm is used. Growth rates are about 0.6 μm/h for GaN (growth temperature  $T_s \sim 800^\circ\text{C}$ ) and 0.1 μm/h for AlN ( $T_s \sim 920^\circ\text{C}$ ). Fig. 1 shows the schematic cross section of the structures we have grown. The HEMT active layers

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consist of a 21-nm  $\text{Al}_{0.28}\text{Ga}_{0.72}\text{N}$  barrier with a 1-nm  $\text{AlN}$  spacer and are covered with a 5-nm  $\text{GaN}$  cap layer. These layers are deposited on a 1.7- $\mu\text{m}$   $\text{GaN}$  buffer grown on a 0.25- $\mu\text{m}$   $\text{AlN}$ /0.25- $\mu\text{m}$   $\text{GaN}$  stress mitigating stack with a 40-nm  $\text{AlN}$  nucleation layer.  $\text{Si}(111)$  substrates, 50 mm in diameter with a resistivity of 10,000  $\Omega\text{cm}$  and a thickness of 380  $\mu\text{m}$  are used for this study. Growth conditions of the buried layers were varied to produce the set of samples of this study. In cases where the first micrometer of the thick  $\text{GaN}$  buffer was grown using the plasma source, a 1.8 sccm nitrogen flow rate was used with a plasma source (Addon RF N50/63) in the 700–720  $^{\circ}\text{C}$  temperature range. In those cases, the growth was performed under Ga-rich conditions with a resulting growth rate close to 0.5  $\mu\text{m}/\text{h}$  at a RF power of 400 W.

The strain state of the as-grown samples is systematically evaluated by measuring the substrate bow with a surface profilometer. Our previous work has confirmed the correspondence between the bow and the curvature generated during the

growth [5]. The stress sensitive low-temperature photoluminescence (PL) energy position in  $\text{GaN}$  [7] has also confirmed this point. Tapping mode atomic force microscopy (AFM) is used to study the surface morphology and to estimate the threading dislocation density (TDD). Plan-view transmission electron microscopy (TEM) confirms that the surface depressions exhibited by AFM are associated with threading dislocations. Furthermore, the width of the asymmetric (302) X-ray diffraction lines is used to measure the average twist of the crystal grains in  $\text{GaN}$  and calculate the dislocation density. Hall effect measurements have been employed to evaluate the quality of the two-dimensional electron gas (2DEG) at the  $\text{AlGaN}/\text{GaN}$  interface of the HEMTs. For this, Van der Pauw devices were made using a two-step lithography process consisting of 0.1  $\mu\text{m}$  depth mesa etching by reactive ion etching and  $\text{TiAlNiAu}$  ohmic contact deposition and annealing.

### 3. Correlation between substrate curvature and layer quality

Fig. 2 shows the dislocation density evaluated with the three different methods. A trend appears: the structures with lower TDDs are grown with more compressive stress (wafers with convex bow). This has a direct effect on the electron transport in the 2DEG, especially on the electron mobility that is sensitive to dislocation scattering. A mobility approaching 2000  $\text{cm}^2/\text{Vs}$  at room temperature is obtained for samples with dislocation densities less than  $5 \times 10^9 \text{ cm}^{-2}$ , which is among the best results reported on silicon substrates. As a consequence, the sheet resistance of the 2DEG clearly decreases with increasing convex bow (Fig. 3) over a relatively large range (350–440  $\Omega/\text{sq}$ ).

Among the parameters that influence the final quality of the layers, we identified the growth temperature of the thick  $\text{GaN}$  buffer layer as one of highest importance. Indeed, our previous studies have shown that a  $\text{GaN}$  buffer layer grown on a 0.25- $\mu\text{m}$ -thick  $\text{AlN}$  suffers an accelerated stress relaxation when grown at 820  $^{\circ}\text{C}$  instead of 800  $^{\circ}\text{C}$ . In the latter case, the residual compressive stress of the  $\text{GaN}$  grown lattice mismatched on the stress relaxed  $\text{AlN}$  layer is not sufficient to compensate the effect of the thermal expansion coefficient mismatch upon cooling down, and cracks are generated. However, the reason for the higher threading dislocation density remains unclear.

|                           |                                     |
|---------------------------|-------------------------------------|
| <b>GaN</b>                | <b>5 nm</b>                         |
| <b>AlGaIn</b>             | <b>21 nm</b>                        |
| <b>AlN</b>                | <b>1 nm</b>                         |
| <b>GaN</b>                | <b>1.7 <math>\mu\text{m}</math></b> |
| Plasma MBE (optional)     |                                     |
| <b>AlN</b>                | <b>250 nm</b>                       |
| <b>GaN</b>                | <b>250 nm</b>                       |
| <b>AlN</b>                | <b>40 nm</b>                        |
| <b>Si (111) substrate</b> |                                     |

Fig. 1. Schematic cross section of the HEMT structures grown on  $\text{Si}(111)$ .

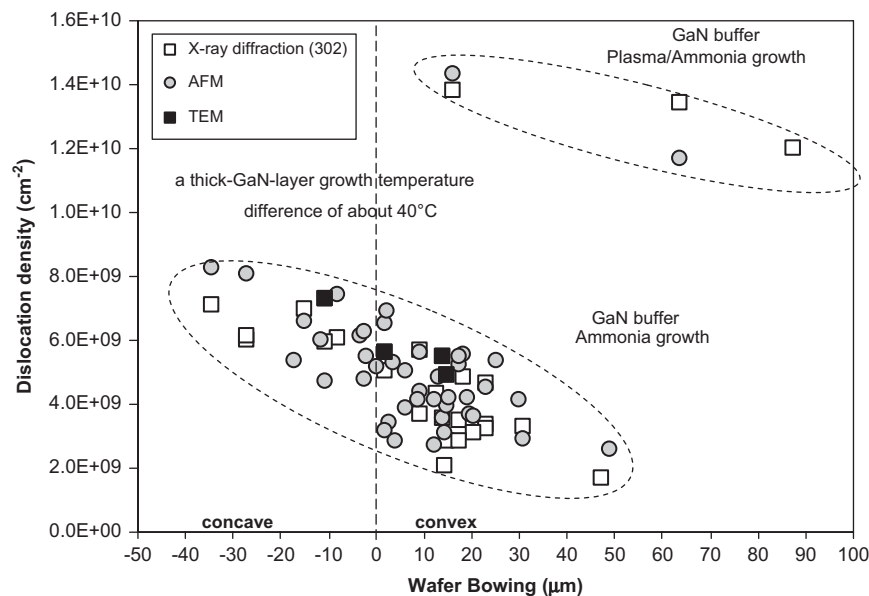


Fig. 2. Correlation between strain state (substrate bow) and dislocation density.

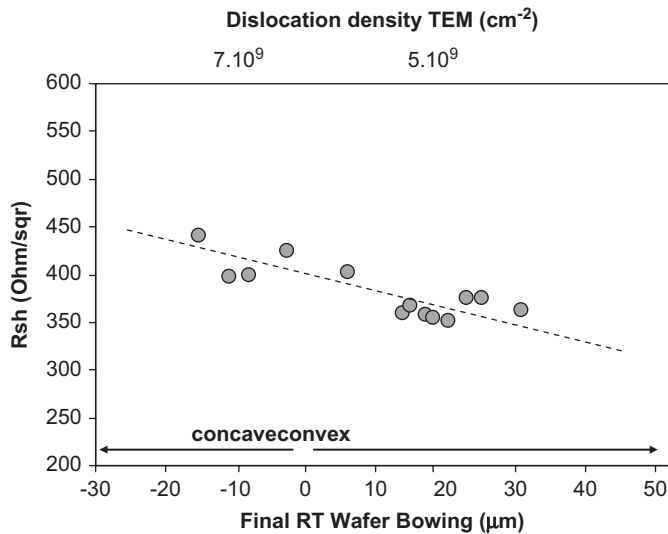


Fig. 3. Room temperature sheet resistance of the 2DEG of AlGaIn/GaN HEMTs versus substrate bow.

#### 4. Dislocation behavior in the buffer layer: role of the temperature

In order to investigate the dislocation behavior, TEM cross-sectional samples were prepared. Fig. 4 shows a picture associated with the diffraction vector  $g=(11\bar{2}0)$ . Under these diffraction conditions, only edge-type dislocations (pure edge  $a$  and mixed type  $a+c$ ) are visible. A very high amount ( $\sim 10^{11}\text{ cm}^{-2}$ ) of threading dislocations propagates from the nucleation layer. The three successive interfaces contribute to the reduction of the dislocation density, but the most effective interface for reducing the dislocation density is clearly the last interface between the thick AlN layer and the main 1.7  $\mu\text{m}$  GaN layer. Just above the thick AlN layer and within the first 200 nm of the thick GaN layer, we observe a region with numerous dislocation loops and where most of the other dislocations are bent. Investigations have shown that a great part of the misfit stress is relaxed within this thin region. This GaN region is initially in a highly compressive stress state owing to the 2.5% lattice mismatch between AlN and GaN, but the stress is relaxed at a high rate. In this region, dislocation interactions are favored, and the compressive stress is gradually relaxed during the entire deposition. The gradual relaxation rate is correlated with decreasing dislocation density with the thickness. Correlation of stress evolution and dislocation behavior has been reported for MOCVD growth as well as for MBE growth [8–11]. For (0001)-oriented layers in the wurtzite crystal structure, the majority of the dislocations are edge type with [0001] line direction. Their Burger vector is of the type  $\frac{1}{3}\langle 11\bar{2}0 \rangle$ , i.e. in the basal plane, and hence these dislocations are not able to glide. As shown in Fig. 4, the lines of these dislocations are inclined with respect to the (0001) growth direction and hence their projection in the growth plane leads to the formation of effective misfit segments. This inclination of the TDs must primarily be due to a climb mechanism and not a dislocation glide process. The inclination angle of dislocation lines with respect to the growth direction decreases with increasing film thickness, and annihilation of dislocations is obvious in the lower part of the main GaN layer. The evolution of the inclination angle is concomitant with the gradual relaxation of the compressive stress because of the decay of the force applied to the dislocation lines. The present observations suggest that at the beginning of the growth of the GaN layer, the stress relaxation process is favored by the noticeable inclination of the dislocations and also

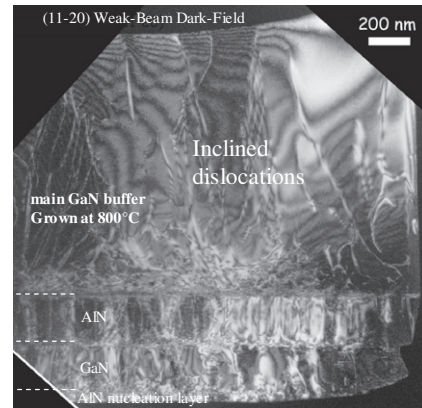


Fig. 4. Cross-sectional TEM image of a structure in which the GaN layer is grown with ammonia only at 800 °C. Only dislocations with an edge component ( $a$  and  $a+c$ ) are visible.

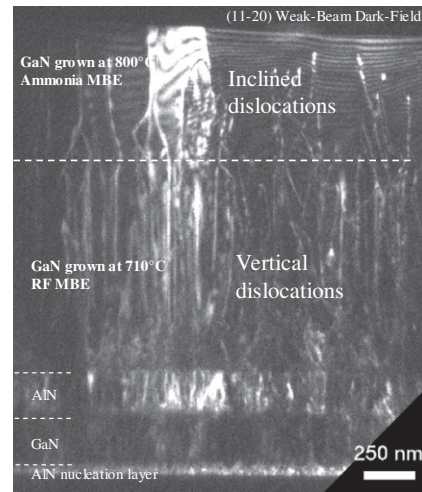


Fig. 5. Cross-sectional TEM image of a structure in which the GaN layer is grown with nitrogen plasma source MBE at 710 °C followed by ammonia source MBE at 800 °C. Only dislocations with an edge component ( $a$  and  $a+c$ ) are visible.

by the fact that this inclination is quite random and favors interactions between the numerous dislocations and consequently enables the nucleation of dislocation loops is enabled. These loops reduce the stress and the number of threading defects as reported by Sahonta et al. [8]. Once this process has sufficiently reduced the compressive stress, fewer dislocations with a smaller inclination are less likely to interact and then the rate of the decay of the dislocation density drops and the compressive stress relaxation rate starts to saturate. Our results suggest that if the relaxation in the early stages of the growth of the GaN buffer is sufficient, the stress is no longer high enough to noticeably incline the still numerous dislocations and to produce their annihilation. As a result, a higher growth temperature may enhance the relaxation process and this faster relaxation rate leads to a thinner compressive portion of the GaN buffer and therefore to a less effective dislocation filtering process in the rest of the structure.

Furthermore, the surface roughness is observed to increase slightly by about 1 nm (RMS for  $2 \times 2 \mu\text{m}^2$  area scans) for the sample grown at lower temperature; then, it is not correlated with the dislocation behavior in the same way as in the MOCVD growth [9]. The surface roughening and the generation of vacancies due to

the limited cracking efficiency of ammonia limit the interest of the growth by ammonia MBE at even lower temperatures. On the contrary, the nitrogen plasma source MBE growth is able to produce good quality layers at noticeably lower temperatures and with a flat growth front thanks to Ga-rich conditions. Fig. 5 shows the edge-type dislocations in a structure where the first micrometer of the 1.7- $\mu\text{m}$ -GaN was grown at 710 °C. As in the case of ammonia MBE, one notices the inclination of the dislocations and the presence of dislocation loops in the buried part of the layer, but the threading dislocations rapidly redirect towards the (0001) direction and propagate vertically until the growth conditions have changed. The fact that most of the previously vertical dislocation lines become inclined again after switching to ammonia MBE at 800 °C confirm the presence of a significant residual compressive stress in the GaN layer. Thus, the lower growth temperature makes it possible to maintain a high residual compressive stress as seen in the noticeable convex bow of the plasma source grown samples in Fig. 2. Furthermore, the limited dislocation interactions explain the less effective dislocation reduction process with a resulting TD density exceeding  $10^{10} \text{ cm}^{-2}$ .

## 5. Conclusion

In this paper, the effect of growth conditions on structural and electrical properties of AlGaIn/GaN HEMT structures grown on (111)-oriented silicon substrates was investigated. A general trend was observed between the final wafer bowing at room temperature and the threading dislocation density. This dislocation density was found to affect the gradual relaxation rate of the compressive stress and the resulting wafer bowing as well as the 2DEG transport properties. We identified the growth temperature of the GaN buffer as a critical parameter for the stress relaxation and consequently for the final quality of the layers and for the wafer curvature. A higher growth temperature leads to a faster rate of compressive stress relaxation and therefore to a lower residual compressive strain at the end of growth, which cannot

compensate the TEC mismatch tensile stress during the post-growth cooling. This faster relaxation rate leads to a thinner compressive portion of the GaN buffer and therefore a less effective dislocation filtering in the rest of the structure. It is suggested that at the beginning of the growth of the GaN buffer, the compressive stress relaxation process is favored by the noticeable inclination of the dislocations and by the interactions between the numerous dislocations enabling the nucleation of dislocation loops. Finally, a mobility approaching  $2000 \text{ cm}^2/\text{Vs}$  at room temperature is obtained for samples with dislocation densities of less than  $5 \times 10^9 \text{ cm}^{-2}$ , which is among the best results reported on silicon. Both the properties of the 2DEG and the structural quality are correlated with the substrate bending, with the very interesting result that the better quality structures are less subject to the risk of generating cracks. On the other hand, the growth of mixed structures with a nitrogen plasma source confirms that the growth temperature plays the most important role for the mechanisms identified for strain relaxation via the dislocation inclination.

## References

- [1] E. Feltin, B. Beaumont, M. Lügt, P. de Mierry, P. Vennéguès, H. Lahrèche, M. Leroux, P. Gibart, *Appl. Phys. Lett.* 79 (2001) 3230.
- [2] F. Semond, Y. Cordier, N. Grandjean, F. Natali, B. Damilano, S. Vézian, J. Massies, *Phys. Status Solidi A* 188 (2001) 501.
- [3] A. Krost, A. Dadgar, G. Strassburger, R. Clos, *Phys. Status Solidi (a)* 201 (2003) R75.
- [4] A. Vescan, J.D. Brown, J.W. Johnson, R. Therrien, T. Gehrke, P. Rajagopal, J.C. Roberts, S. Singhal, W. Nagy, R. Borges, E. Piner, K. Linthicum, *Phys. Status Solidi (c)* 0 (2002) 52.
- [5] Y. Cordier, N. Baron, F. Semond, J. Massies, M. Binetti, B. Henninger, M. Besendahl, T. Zettler, *J. Crystal Growth* 301–302 (2007) 71.
- [6] Y. Cordier, F. Semond, J. Massies, M. Leroux, P. Lorenzini, C. Chaix, *J. Crystal Growth* 301–302 (2007) 434.
- [7] H. Lahrèche, M. Leroux, M. Lügt, M. Vaille, B. Beaumont, P. Gibart, *J. Appl. Phys.* 87 (2000) 577.
- [8] S.-L. Sahonta, M.Q. Baines, D. Cherns, H. Amano, F.A. Ponce, *Phys. Status Solidi (b)* 234 (2002) 952.
- [9] A.E. Romanov, J.S. Speck, *Appl. Phys. Lett.* 83 (2003) 2569.
- [10] S. Vézian, F. Natali, F. Semond, J. Massies, *Phys. Rev. B* 69 (2004) 125329.
- [11] A.L. Corrión, C. Poblénz, F. Wu, J.S. Speck, *J. Appl. Phys.* 103 (2008) 093529.